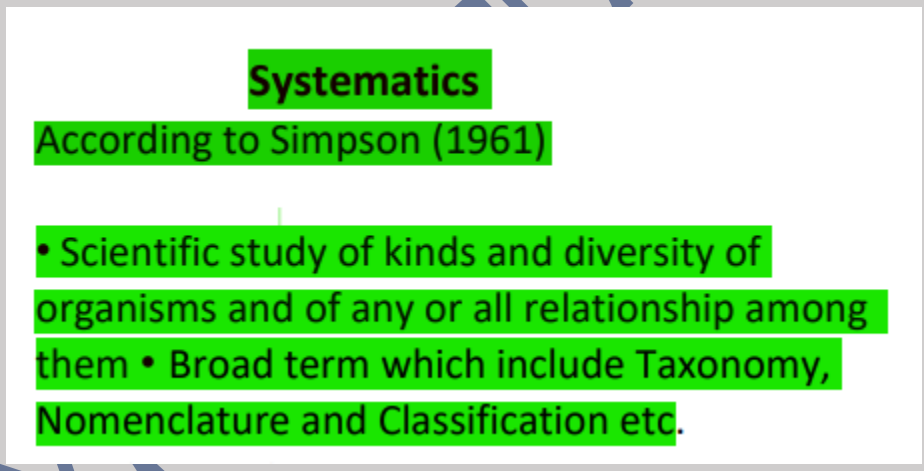
Zoo103 Current Questions 2022

By M Zaman

Only Current **IMPORTANT Subjective** Questions, which are not included in any recent file, so I just gather here those question for your ease.



Systematics

According to Simpson (1961)

• Scientific study of kinds and diversity of organisms and of any or all relationship among them • Broad term which include Taxonomy, Nomenclature and Classification etc.

Construction of Phenogram

- A tree diagram depicting taxonomic relationships among organisms based on overall similarity of many characteristics without regard to evolutionary history

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✓ Catalogues & Check lists

A catalogue is an index to taxa arranged in such a way to give a vivid picture of refere nces for both zoological and nomenclature purpose:

✓ Monographs:

A comprehensive systematic study of a particular taxonomic group.

• Always taxon-based, not geography-based.

• Its preparations require more complete material than other types of publications.

• 3-5 years for completion. • Monographs try to review all the facts about a taxonomic group.

•A Monograph is a complete global account of a taxon of any rank – family, genus or species at a given time.

• Book format

Faunal/Floral Work

It deals with detail description faunal/Floral work of particular region.

• It may include information on geographical distribution.

• Also include quantitative data and ecological comments.

• Only experience specialists.

• Example: Altas of scale insects of North America

Module# 152 Taxonomic publications

- No taxonomic study is complete until its results are published.
- Advances in any area of science is made through publications.
- Taxonomic publications range from short description of species (may be one page) to monograph or handbook (several volumes)

- A parasitoid is an organism that has young that develop on or within another organism (the host), eventually killing it.
- Parasitoids have characteristics of both predators and parasites.
- Parasitoids are usually smaller than their selected host
- Pathogen is an infectious agent that feed on host body and cause damage to its host.
- Bacteria
- Viruses
- Protozoon

- Others may seprate couplets **Types of Keys**
- Indented Key

Construction of Identification Key

- Bracket Key
- Pictoral Key
- Polyclave Key

• Discovery of the fact that previously morphospecies were not reproductively isolate and did not qualify as a separate species hence…sub species

Taxon

• Defined by simpson (1961) as 'a group of real organisms recognized as a formal unit at any level of hierarchical classification'

Taxonomy of Green Lace wings

Phylum	Arthropoda
Class	Insecta
Family	Chrysopidae
Genus	Chrysoperla
Species	C. carnea

Allopatric Speciation

- The most common form of speciation
- Occurs when populations of a species become geographically isolated.

Examples

- Speciation is the Galápagos finch

17. Sympatric Speciation

- A speciation in which a species is evolved from a single common ancestral species while inhabiting the same geographical range.
- Share the same habitat but become reproductively isolated from each other.

- More common in plants

- Batesian mimicry is a type of mimicry in which platable animal mimics the bad-tasting or noxious prey so that predator may not distinguish b/e them.
- After several attempts of distasteful encounters predators learn to avoid such appearance.
- It is called Batesian mimicry in honor of English naturalist Henry W. Bates who published an explanation of this mimicry in 1862.

• The lateral line system functions as a sensory organ to sense the disturbances in the nearby environment.

Phylum Echinodermata

• They are exclusively marine and are among the most common and widely distributed marine animals.

• Symmetry usually radial, nearly always pentamerous.

• Body shape globular, star-like, spherical, discoidal, or elongated.

• Water -vascular system of coelomic origin, including podia or tube feet for locomotion and usually with a madreporite.

44. Ecological Variations

A. Habitat Variation
B. Variations induced by climatic conditions
C. Host determined variations
D. Density dependent variations
E. Allometric variations
F. Neurogenic color variations

Module# 180 Lac Insect

Kingdom	Animalia
Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Kerriidae
Genus	Kerria
Species	*Kerria lacca*

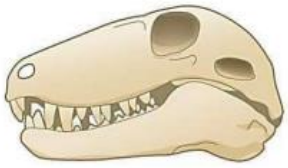
- Lac insect live as parasite on sap of certain tress and shrubs and shows sexual dimorphism
- *Babul (Acacia arabica)*
- *Khair (Acacia catcchu)*
- Its life is completed in 6 months
- Two crops can be obtained



Anapsid skull (Turtles)



Diapsid skull (Reptiles including birds)



Synapsid skull (Mammals)

Internal morphology

- Internal anatomy provides Taxonomic Characters for all groups of higher animals. • The extend to use vary from group to group.
- The skull of routinely used in classification of higher mammals. • Teeth and bones of fossils and other hard parts.

https://www.google.com/search?q=evolution+in+skulls+of+vertebrates

Mimicry

- Mimicry refer to similarities between species.
- Mimicry is a form of symbiosis where species may mimic colours, patterns or even behaviours for one of the two reasons:

A. Either to be able to get closer to unsuspecting prey

B. To gain protection by attaining similarity with predators or poisonous species.

- Mostly in mimictic relationship advantage is one side.

Types of Mimicry

- Batesian mimicry
- Mullerian mimicry
- Wasmannian mimicry
- Peckhmian mimicry
- Self-mimicry

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

Remember me in your Precious Prayers

Jazak Allah

M Zaur